DRIFTLESS — Configuration & Dependency Specifications

1. Python Dependencies (`dashboard/requirements.txt`)

```
Django≥4.2,<5.0
djangorestframework≥3.15
django-cors-headers≥4.3
whitenoise≥6.6
python-dotenv≥1.0
gunicorn≥21.2
sqlparse≥0.5
```

Detailed Dependency Explanations:

1. **`Django≥4.2,<5.0`**: Long-Term Support (LTS) release of the Django web framework. Provides the core ORM, database migrations, model definitions, session management, template rendering, and administrative back office.
2. **`djangorestframework≥3.15`**: The standard toolkit for REST APIs in Django. Powers serializer validation, token-based authentication (`rest_framework.authtoken`), view decorators (`@api_view`), permission classes, and throttling engines.
3. **`django-cors-headers≥4.3`**: Middleware that attaches Cross-Origin Resource Sharing (CORS) headers to HTTP responses. Essential for enabling the Chrome extension runtime (`chrome-extension:// ...`) to make fetch calls to the Django backend.
4. **`whitenoise≥6.6`**: Static file serving engine. Binds directly to the WSGI pipeline, enabling Gunicorn to serve production static assets (CSS, logos, themes) with Brotli/Gzip compression and permanent caching headers without requiring Nginx.
5. **`python-dotenv≥1.0`**: Automatically discovers and parses key-value pairs from `.env` files into Python's `os.environ`, decoupling sensitive secrets and deployment flags from code.
6. **`gunicorn≥21.2`**: Production-grade WSGI HTTP server designed for Unix/Linux container environments (Render, Heroku, Docker).
7. **`sqlparse≥0.5`**: Non-validating SQL parser required internally by Django's database management engine to format and execute migration scripts.

2. Environment Variables Specification (`dashboard/.env` & `.env.example`)

Variable Name	Default in Code	Purpose	Production Requirement
`SECRET_KEY`	`'driftless-dev-secret-change-in-production-pleaseee'`	Cryptographic signing key for sessions, CSRF tokens, and passwords	**MANDATORY:** Must be set to a secure 50+ character random string in production
`DEBUG`	`True`	Toggles Django debug traceback pages and query logging	**MANDATORY:** Must be set to `False` in production
`ALLOWED_HOSTS`	`*`	Comma-delimited list of host/domain names that this Django site can serve	Set to production domain (e.g. `driftless.onrender.com,localhost`)
`CORS_ALLOW_ALL_ORIGINS`	`True`	Enables open CORS access	Can remain `True` or restrict to specific origins and Chrome extension regex
`CSRF_TRUSTED_ORIGINS`	Ngrok, Cloudflare, Localhost list	Comma-delimited origins trusted for POST requests	Include production domain (e.g. `https://driftless.onrender.com`)
`SECURE_SSL_REDIRECT`	`False`	Forces all HTTP requests to be redirected to HTTPS	Set to `True` when running behind SSL-terminating reverse proxies

3. Extension Manifest Specification (`extension/manifest.json`)

```
{
  "manifest_version": 3,
  "name": "Driftless - Focus & Wellbeing Companion",
  "version": "1.1",
  "description": "Digital wellbeing & distraction tracker with per-site limits, focus mode,
and cloud sync.",
  "icons": {
    "16": "icons/icon-16.png",
    "48": "icons/icon-48.png",
    "128": "icons/icon-128.png"
  },
  "action": {
    "default_popup": "popup.html",
    "default_icon": {
      "16": "icons/icon-16.png",
      "48": "icons/icon-48.png",
      "128": "icons/icon-128.png"
    }
  },
  "permissions": [
    "tabs",
    "storage",
    "alarms",
    "notifications",
    "idle",
    "declarativeNetRequest",
    "declarativeNetRequestWithHostAccess",
    "declarativeNetRequestFeedback"
  ],
  "host_permissions": ["<all_urls>"],
  "web_accessible_resources": [
    {
      "resources": ["blocked.html", "theme.css", "icons/*"],
      "matches": ["<all_urls>"]
    }
  ],
  "background": {
    "service_worker": "background.js"
  }
}
```

4. Build & Development Scripts

4.1. Extension Packager: `scripts/build_extension.py`

- **Execution:** `python scripts/build_extension.py`

- **Behavior:**
  1. Resolves `ROOT_DIR / 'extension'`.
  2. Unlinks any pre-existing `driftless-extension.zip`.
  3. Recursively gathers all files in `extension/`, excluding hidden OS/Git files.
  4. Generates a clean, deflated zip file (`driftless-extension.zip`) ready for direct distribution or uploading to `chrome://extensions`.

4.2. Shell Alternative: `scripts/build_extension.sh`

- A Bash equivalent for Unix/macOS environments using standard `zip -r`.

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5. Deployment Configurations

5.1. `dashboard/Procfile`

```
web: gunicorn driftless.wsgi:application --bind 0.0.0.0:$PORT
```

Declares the process command for PaaS platforms like Render or Heroku to launch Gunicorn bound to the dynamically assigned `$PORT`.

5.2. `dashboard/render.yaml`

```
services:
  - type: web
    name: driftless-dashboard
    env: python
    buildCommand: "pip install -r requirements.txt && python manage.py migrate && python
manage.py collectstatic --noinput && python manage.py seed_demo"
    startCommand: "gunicorn driftless.wsgi:application"
    envVars:
      - key: PYTHON_VERSION
        value: 3.10.12
      - key: DEBUG
        value: "False"
      - key: SECRET_KEY
        generateValue: true
```

Infrastructure-as-code specification for Render.com that automates dependency installation, database migration, static collection via WhiteNoise, demo account seeding, and Gunicorn startup.

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6. Git Ignore Directives (`.gitignore`)

The repository includes `.gitignore` rules that protect the codebase against accidental leakage of:
- Virtual environments (`venv/` , `env/` , `.venv/`)
- Compiled Python bytecode (`__pycache__/` , `*.py[cod]`)
- SQLite database binaries (`*.sqlite3`)
- Environment secret files (`.env`)
- Static collection dumps (`staticfiles/`)
- Operating system cache files (`Thumbs.db` , `.DS_Store`)